

## Scalability & Future Plan

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|----------------------|--------------|
| <b>Date</b>          | 05 July 2026 |
| <b>Team ID</b>       |              |
| <b>Project Name</b>  | Smart Lender |
| <b>Maximum Marks</b> | 1 Mark       |

### Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

### Current System Limitations:

| S.No | Limitation                             | Impact                                                                                                                                                       | Priority to Address (High / Medium / Low) |
|------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| 1    | SQLite Concurrent Write Constraints    | Under multiple concurrent user requests, SQLite's database-level lockups will result in failed database writes (locks) and application exceptions.           | High                                      |
| 2    | Pickled Local Model File Storage       | Model binaries (rdf.pkl and scale1.pkl) are stored on the local file system. This prevents serverless horizontal scaling and makes model updating difficult. | Medium                                    |
| 3    | Lack of User Authentication & Security | Anyone can navigate to the /history tab and purge database logs without identification, posing significant data security and integrity risks.                | High                                      |

### Scalability Plan:

| S.No | Scalability Aspect | Current State                                                               | Proposed Upgrade / Solution                                                                                             |
|------|--------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| 1    | User Load          | Runs on a single-instance development Flask server (synchronous, blocking). | Deploy using a multi-worker production WSGI server (Gunicorn) containerized with Docker, behind an Nginx reverse proxy. |
| 2    | Data Storage       | Uses a local SQLite                                                         | Migrate to a managed cloud database system (e.g.,                                                                       |

|   |             |                                                                                  |                                                                                                                                                 |
|---|-------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
|   |             | file<br>( <code>smart_lender.db</code> ).                                        | PostgreSQL or Amazon RDS) to support high concurrent transactions and connection pooling.                                                       |
| 3 | Performance | Python synchronous model execution directly within the Flask web request thread. | Decouple the ML inference engine to a dedicated microservice (using FastAPI or BentoML) with caching (e.g., Redis).                             |
| 4 | Security    | Open routes without authentication or rate limiting.                             | Implement Flask-Login or OAuth2 for authentication, role-based access control (RBAC), and integration of Flask-Limiter to prevent request spam. |

### Future Roadmap:

| Phase          | Planned Feature / Enhancement                           | Target Timeline | Expected Impact                                                                                               |
|----------------|---------------------------------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------|
| <b>Phase 1</b> | Implement User Authentication & Authorization (RBAC).   | 2-4 Weeks       | Secures the system; ensures only authorized underwriters can approve loans or delete prediction histories.    |
| <b>Phase 2</b> | Database Migration to PostgreSQL & Dockerization.       | 1-2 Months      | Prepares the application for cloud deployment (e.g., Render/AWS) with scale-out database support.             |
| <b>Phase 3</b> | Decouple Inference Engine to FastAPI & Auto-Retraining. | 3-4 Months      | Decouples model inference from the UI server, lowering latency and enabling live updates to models.           |
| <b>Phase 4</b> | Integrate Explainable AI (XAI) using SHAP and LIME.     | 6+ Months       | Displays feature importance plots to applicants, providing transparent explanations for approvals/rejections. |